

Sequences of real numbers

Remarkable sequences:

$$\cdot C_n = \left(1 + \frac{1}{n}\right)^n \nearrow e = 2,7182$$

$$\cdot E_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!} \nearrow e$$

$$\cdot \gamma_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n \searrow \gamma \approx 0,57 \text{ (Euler's constant)}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right)$$

$$H_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} \text{ (Harmonic sequence)}$$

$$\lim_{n \rightarrow \infty} H_n = \lim_{n \rightarrow \infty} (\gamma_n + \ln n) = \gamma + \infty = +\infty$$

Pa1 Calculate:

$$a) \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}\right)$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{2} + \dots + \frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{2n} - \ln(2n)\right)$$

$\underbrace{\hspace{15em}}_{\gamma_{2n}}$

$$- \underbrace{\left(1 + \frac{1}{2} + \dots + \frac{1}{n} - \ln n\right)}_{\gamma_n} + \ln(2n) - \ln n$$

$$\lim_{n \rightarrow \infty} (x_{2n}^2 - x_n^2 + \ln(\frac{2n}{n})) = x^2 - x^2 + \ln 2 = \ln 2$$

$$b) \lim_{n \rightarrow \infty} (e^{1 + \frac{1}{2} + \dots + \frac{1}{n+1}} - e^{1 + \frac{1}{2} + \dots + \frac{1}{n}})$$

$$\lim_{n \rightarrow \infty} e^{1 + \frac{1}{2} + \dots + \frac{1}{n}} \left(\frac{e^{\frac{1}{n+1}} - 1}{\frac{1}{n+1}} \right) \cdot \frac{1}{n+1}$$

$$\lim_{n \rightarrow \infty} (e^{x_n + \ln n} \cdot \frac{1}{n+1})$$

$$\lim_{n \rightarrow \infty} (e^{x_n} \cdot \frac{e^{\ln n}}{n+1}) = \lim_{n \rightarrow \infty} e^{x_n} \cdot \frac{n}{n+1} = e$$

Important criteria for limits of sequences

I Stolz - Cesaro Th. ($\frac{\infty}{\infty}$)

Let $(a_n)_{n \geq 1}$, $(b_n)_{n \geq 1}$ be 2 sequences of real numbers with the following properties:

- 1) $(b_n)_{n \geq 1}$ is strictly monotonic and unbounded
- 2) $\exists \lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \ell \in \overline{\mathbb{R}}$

Then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \ell$

Pr 2 Calculate:

$$a) \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \dots + \frac{1}{n}}{\ln n}, n \geq 2$$

$$1) \ln n \nearrow$$

$$2) \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \dots + \frac{1}{n} + \frac{1}{n+1} - 1 - \frac{1}{2} - \dots - \frac{1}{n}}{\ln(n+1) - \ln n}$$

$$\lim_{n \rightarrow \infty} \frac{1}{(n+1) \cdot \ln\left(\frac{n+1}{n}\right)} = \lim_{n \rightarrow \infty} \frac{1}{\ln\left(\frac{n+1}{n}\right)^{n+1}} = \frac{1}{\ln e}$$

$$= 1$$

$$\lim_{n \rightarrow \infty} \frac{1 + \frac{1}{2} + \dots + \frac{1}{n}}{\ln n} = 1$$